

Dear Editor,

I am pleased to submit our manuscript entitled “Surface Roughness and Standard Choice Drive Gear Bending Fatigue Life Prediction Divergence: A Full-Factorial Variance Decomposition” for consideration for publication in the *International Journal of Fatigue*.

What this paper does. We apply a full-factorial ANOVA framework to decompose the total variance in gear tooth bending fatigue life prediction into six methodological sources. Holding geometry, material, and loading constant, we systematically vary the choices an engineer must make: which standard (ISO 6336, AGMA 2001, FKM), which surface roughness grade ($R_z = 4, 15, 50 \mu\text{m}$), which S–N curve (MatWeb, MMPDS), which mean-stress correction (Goodman, Gerber, Morrow, SWT), and so on. The analysis reveals that mean-stress correction method (39.0%) and standard choice (34.5%) together account for nearly three-quarters of all prediction divergence—a finding that contradicts the common assumption that mean-stress effects are minor for gear bending at $R = 0$. All formulas in the fatigue model have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.).

Why IJ Fatigue. The paper addresses a core question in fatigue life prediction—how much does methodological choice matter relative to physical uncertainty?—using a quantitative variance-decomposition framework that has been applied in other fields but is new to gear fatigue. The analysis is purely computational (no experimental data), uses only open-source tools and publicly available material databases, and is fully reproducible. We believe the methodology and findings are directly relevant to the journal’s readership in fatigue life prediction, standards development, and uncertainty quantification.

Novelty and significance. To our knowledge, this is the first study to deploy a controlled full-factorial ANOVA to rank the sources of method-induced divergence in gear bending fatigue life prediction. Prior work has compared individual factors in isolation (e.g., Goodman vs. Gerber, or ISO vs. AGMA) but has not systematically mapped the full interaction structure. The finding that mean-stress correction is the largest source of variance at $R = 0$ —contradicting engineering intuition—is both methodologically novel and practically significant.

The manuscript has not been published and is not under consideration elsewhere. All authors have approved the submission.

AI assistance disclosure. Language editing and code development for this work were assisted by large language models (Claude, Anthropic; Hermes Agent). All numerical results were generated by executing the open-source Python pipeline provided with the submission. Every formula in the fatigue model has been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.). The author takes full responsibility for the scientific content.

Sincerely,

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